# Spencer Presley

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## EXPERIENCE

**CrunchAtlas — Software Engineer** | Mar 2025 – Present

*Local-first, air-gapped AI security platform for critical infrastructure. Sole author of the systems below.*

- **Created the company's agent platform from scratch.** A shared harness supplies model construction, retries, structured-output repair, context management, tool guardrails, per-task budgets, LangGraph-checkpointed run state, and typed lifecycle events; each agent adds only its domain prompt, tools, state, and graph.
- **Kept long agent runs inside a bounded context.** Before every llama.cpp call the harness tokenizes the exact request and refuses anything that won't fit; it condenses a filling run to its objective, completed work, and next steps, moves oversized tool results to searchable files, and drops a truncated turn before memory or tools see it.
- **Gave understaffed critical-infrastructure teams on-demand packet forensics** through the platform's most-used agent. It drives tshark, Zeek, and foremost from an IDS alert through protocol reconstruction, IOC extraction, file carving, and timeline analysis, then delivers a cited report and triage verdict.
- **Made the platform investigate on its own**, completing nearly 17,000 PCAP investigations in its first two months with no analyst click. A background reconciler decides which AI work should exist and queues it, recording its progress durably so a crash loses nothing already queued.
- **Shipped PurpleHaze**, an autonomous penetration-testing product covering reconnaissance, exploitation, web, and credential testing. Its Metasploit tool keeps modules, options, and Meterpreter sessions alive across model turns and context compaction, so a full engagement ends in OWASP and NIST reports.
- **Replaced fixed LLM concurrency with token-budgeted GPU scheduling.** Five fixed 131k-token llama.cpp slots became one 655k-token KV-cache pool that holds any job mix whose declared context fits, from 7 concurrent 128k-context agents to 160 4k-token jobs, sized byte-exact against llama.cpp's KV cache.
- **Kept long-running GPU and sandbox jobs independently cancellable** and returned in-flight work to a terminal state after crashes or hangs. The Postgres scheduler combines atomic `SELECT FOR UPDATE SKIP LOCKED` claims, interactive-first fair-share ordering, per-run process isolation, heartbeats, and process-tree cancellation.

**Horn Point Laboratory — Software Engineering Intern** | May 2024 – Aug 2024

- **Made an NSF-funded Chesapeake Bay salinity corpus searchable** through a RAG chatbot (FAISS, OpenAI, streaming FastAPI/LangChain); presented at the MIT Undergraduate Research Technology Conference.

## PROJECTS

**docq** — *Python, SQLite/FTS5, Ollama*

- Created a cited retrieval engine that lets coding agents search a long reference instead of loading all of it. On a leakage-screened 266-case eval, hybrid BM25/vector retrieval reached 0.67 hit@1 and beat either channel alone; a instruction-based query-rewrite A/B raised it to 0.74 (p=0.0045).

**BitNet b1.58 Reimplementation** — *PyTorch, ternary-weight LM, byte-level, Android*

- Trained ternary-weight ({−1, 0, +1}) language models from scratch at three sizes up to 300M parameters, then deployed one to an Android phone and profiled on-device CPU, memory, and energy.

## HONORS

- **1st Place / Best Hack — Bitcamp 2024**, UMD (largest US East Coast hackathon) — TermBook
- **Best Educational Hack & 2nd Overall — HackUMBC 2024** — Testif.AI

## SKILLS

- **AI & agents:** Agent orchestration, tool calling, context engineering, guardrails, prompt-injection defense, model evaluation, model fine-tuning, RAG, LLM serving, LangChain, LangGraph, llama.cpp, llama-swap, Ollama, OpenAI and Anthropic APIs (prompt caching), Pydantic, PyTorch, NumPy
- **Languages:** Python, Rust, TypeScript, SQL, Bash
- **Backend & infra:** Django (DRF, Ninja), FastAPI, PostgreSQL, Redis, Docker, Linux, Terraform, GitHub Actions
- **AWS:** EC2, S3, CloudFront, ALB, WAF, Route 53, IAM, RDS, ECR, Lambda, SES, Amplify, SSM, CloudWatch
- **Frontend:** React, TanStack Query, OpenAPI codegen (orval, Zod), Vite

## EDUCATION

**Salisbury University** — B.S. Computer Science, Minor in Mathematics | May 2025